Many-one reductions preserve computable enumerability and computability

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-computability-computability-theory/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: both closure directions are fully argued in-span; the parenthetical alternative for claim (2) is the author's own and is delivered verbatim

Editorial note. The label prop:reduce names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Characterisation of computability by c.e. of a set and its complement. Same unit, \ollabel{thm:ce-comp} (source lines 678-682). Second claim: c.e.-ness of A and its complement, obtained from B's computability through the reduction, yields computability of A.

Source reference. Original theorem thm-ce-comp: sources/olp-computability-computability-theory/pinned-original-source.tex, source lines 678–682. Prerequisite pointer invoked in the counted proof; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Proposition 0.1. *Let A and B be any sets, and suppose $A \leq_m B$.*

1. *If B is computably enumerable, so is A.*
2. *If B is computable, so is A.*

Proof. Let f be a many-one reduction from A to B. For the first claim, just check that if B is the domain of a partial function g, then A is the domain of $\circ fg$:

$$\begin{aligned} x \in A &\text{ iff } f(x) \in B \\ &\text{iff } g(f(x))\downarrow. \end{aligned}$$

For the second claim, remember that if B is computable then B and $\overline{B}$ are computably enumerable (thm-ce-comp). It is not hard to check that f is also a many-one reduction of $\overline{A}$ to $\overline{B}$, so, by the first part of this proof, A and $\overline{A}$ are computably enumerable. So A is computable as well by thm-ce-comp. (Alternatively, you can check that $\chi A = \circ f \chi B$; so if χB is computable, then so is χA.) $\square$